CIP-B106

DFIT-COHORT-11

Case Study 2 - Android 10 Pixel 3 Evidence Investigation

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Executive Summary

This report presents the forensic examination of a logical extraction from a Google Pixel 3 running Android 10. Integrity verification was performed using SHA256 hashing before analysis commenced.

The investigation identified device configuration information, user accounts, Wi-Fi and Bluetooth artefacts, and evidence of application usage. Examination of WhatsApp databases recovered contacts, message content, timestamps, and voice/video call records. Additional analysis of Google Maps, Google Photos, and Android Contacts databases revealed location information, media activity, and stored contact details.

The recovered artefacts demonstrate how Android system and application databases can be used to reconstruct user activity and provide valuable evidence during a mobile forensic investigation.

1) Evidence & Method

Evidence log

Item	Value
Case ID	CIPB106-Android10-C11/26/DFIT/17292
Evidence Source	Pixel 3 Android 10 Course Image
Acquisition Type	Logical Copy
Acquisition Method	TWRP-assisted extraction
Examiner	Sandrin Wihbedimbom
Date Examined	11 October 2026
Tools	SQLite Browser, Notepad++, Python scripts etc

Verifying the hash

```
┌──(root㉿icdfa)-[/mnt/hgfs/android]
└─# sha256sum 'Pixel 3.zip'
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  Pixel 3.zip
```

Identifying the Device Build Information

```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# head -100 ./product/build.prop

# begin common build properties
# autogenerated by build/make/tools/buildinfo_common.sh
ro.product.build.date=Tue Dec 17 19:34:12 UTC 2019
ro.product.build.date.utc=1576611252
ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
ro.product.build.id=QQ1A.200105.003
ro.product.build.tags=release-keys
ro.product.build.type=user
ro.product.build.version.incremental=6079943
ro.product.build.version.release=10
ro.product.build.version.sdk=29
ro.product.product.brand=google
ro.product.product.device=blueline
ro.product.product.manufacturer=Google
ro.product.product.model=Pixel 3
ro.product.product.name=blueline
# end common build properties

#
# from device/google/crosshatch/product.prop
#

persist.rild.nitz_plmn=
persist.rild.nitz_long_ons_0=
persist.rild.nitz_long_ons_1=
persist.rild.nitz_long_ons_2=
persist.rild.nitz_long_ons_3=
persist.rild.nitz_short_ons_0=
persist.rild.nitz_short_ons_1=
persist.rild.nitz_short_ons_2=
persist.rild.nitz_short_ons_3=
DEVICE_PROVISIONED=1

# Set network mode to Global by default and no DSDS/DSDA
ro.telephony.default_network=10
```

Examination of the extracted build properties identified the device as a Google Pixel 3 (codename: blueline). The device was running Android 10 (SDK 29) with build QQ1A.200105.003 and security patch level 2020-01-01. The build fingerprint was recorded as google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys.

User Accounts

Analysis of the Android Account Manager database (accounts_ce.db) identified multiple registered accounts associated with the device. The primary Google account recovered was thisisdfir***@gmail.com (com.google). Additional application-specific accounts were identified for WhatsApp, Telegram, Signal, TextNow, Skype, Viber, Twitter, Messenger, TikTok and other applications. The presence of these records confirms that the applications had been configured and associated with user account information on the device.

```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# sqlite3 ./data/system_ce/0/accounts_ce.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .tables
accounts      android_metadata      authtokens      extras
sqlite> SELECT * FROM accounts;
```

_id	name	type	password
1	thisisdfir@gmail.com	com.google	aas_et/AKppINYHk3843SOlirhnRUtCveD5dMEkDbJE-UM6ATDHh15WhYU1lcZNba5nhbrKUmsSReDEDBsCeGL7JPU4q-Yg4xnvw0l3EIJFTafNRpTQp9VWTc60f96ZOmBpTtd9rcFCO31RD2qnXx2XDpWJ7u0jnSIu78gwiCGPaUpaegazIAQxH1Pa-1h0VwGXElenMabM5yUWpAIhzpatx3xMFoA=
2	'858233690'	org.telegram.messenger	
3	Signal	org.thoughtcrime.securesms	NULL
4	imo HD	com.imo.android.imous	NULL
5	TikTok	com.zhiliaoapp.musically	NULL
6	Messenger	com.facebook.messenger	NULL
7	TDfir	com.twitter.android.auth.login	NULL
8	WhatsApp	com.whatsapp	NULL
10	thisisdfir	com.silentcircle.account	dummyPassword
11	Skype	com.skype.raider	NULL
12	TextNow	com.enflick.android.TextNow.account	
13	'+19195794674'	com.viber.voip	961ea18e43b4a2b35216e208d9414212e4b699c3
14	Duo	com.google.android.apps.tachyon	NULL

User Profiles

Examination of userlist.xml identified multiple user profiles configured on the device. User ID 0 represents the primary device user, while User ID 11 represents an additional user profile. No device owner profile was identified, as the deviceOwnerUserId value was set to -10000. The presence of multiple user identifiers indicates the device supported more than one user environment.

```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# find . -name "userlist.xml"
./data/system/users/userlist.xml
./sbin/.magisk/mirror/data/system/users/userlist.xml

┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# cat ./data/system/users/userlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
    <user id="11" />
</users>
```

Wi-Fi Evidence

```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# grep -i "SSID" ./data/misc/wifi/WifiConfigStore.xml

┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# head -50 ./data/misc/wifi/WifiConfigStore.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="2" />
<Integrity>
<byte-array name="EncryptedData" num="48">e71394aa95f2d767a1c46e787ca99c0478e12467241553a8551005969fd38294e8a4082529d69a15d5bc49f28ca21a54</byte-array>
<byte-array name="IV" num="12">18234dad790da5cb3ba8eceb</byte-array>
</Integrity>
<NetworkList />
<MacAddressMap>
<map name="MacMapEntry">
<string name="&quot;CcookiesDcastleR5&quot;WPA_PSK">ae:f6:c7:a8:a5:c2</string>
</map>
</MacAddressMap>
<PasspointConfigData>
<long name="ProviderIndex" value="0" />
</PasspointConfigData>
</WifiConfigStoreData>

┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# grep -i "WifiConfiguration" ./data/misc/wifi/WifiConfigStore.xml
```

Examination of WifiConfigStore.xml identified wireless network configuration artefacts. The

recovered data contained references to a previously configured wireless network (CCoolDCookiesDcastler5) together with associated network configuration information. Portions of the Wi-Fi configuration data were stored in encrypted form, limiting direct interpretation of all saved wireless network parameters.

Bluetooth

Examination of the Bluetooth database (bluetooth_db) identified Bluetooth configuration records. Analysis of the metadata table recovered the Bluetooth address B4:EC:02:73:FF:93. A total of two records were present within the metadata table, confirming Bluetooth-related activity and configuration data were stored on the device.

```
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT address FROM metadata;
     address
_________________
B4:EC:02:73:FF:93
LocalStorage
sqlite> SELECT address, migrated FROM metadata;
     address         migrated
_________________   ________
LocalStorage               1
B4:EC:02:73:FF:93          0
sqlite> SELECT COUNT(*) FROM metadata;
COUNT(*)
________
       2
```

Acquisition Limitations

The supplied Pixel 3 evidence was acquired through a logical extraction after TWRP root access. This acquisition method enabled collection of application and system artefacts but does not establish that the original device owner rooted the device. Root access formed part of the training acquisition procedure and should not be interpreted as evidence of historical user activity.

2) Application Inventory (Task B)

```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# grep -c "<package " ./data/system/packages.xml
291

┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# grep "<package " ./data/system/packages.xml > packages.txt

┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# head packages.txt
    <package name="com.google.omadm.trigger" codePath="/product/priv-app/OemDmTrigger" nativeLibraryPath="/product/priv-app/OemDmTrigger/lib" p
rimaryCpuAbi="arm64-v8a" publicFlags="537411141" privateFlags="524296" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="1" sharedUse
rId="1001" isOrphaned="true">
    <package name="com.google.android.carriersetup" codePath="/product/priv-app/CarrierSetup" nativeLibraryPath="/product/priv-app/CarrierSetup
/lib" publicFlags="537443909" privateFlags="524296" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="29" userId="10157" isOrphaned="
true">
    <package name="com.skype.raider" codePath="/data/app/com.skype.raider-T-lF3_kJap3DnQQzr3UxEw==" nativeLibraryPath="/data/app/com.skype.raid
er-T-lF3_kJap3DnQQzr3UxEw==/lib" primaryCpuAbi="arm64-v8a" publicFlags="810040900" privateFlags="0" ft="16ff2454aa8" it="16ff245509a" ut="16ff2
45509a" version="1250088185" userId="10236" installer="com.android.vending" categoryHint="4">
    <package name="com.android.cts.priv.ctsshim" codePath="/system/priv-app/CtsShimPrivPrebuilt" nativeLibraryPath="/system/priv-app/CtsShimPri
vPrebuilt/lib" primaryCpuAbi="arm64-v8a" secondaryCpuAbi="armeabi-v7a" publicFlags="-1341604287" privateFlags="8" ft="11e8f7d4c00" it="11e8f7d4
c00" ut="11e8f7d4c00" version="28" userId="10073" isOrphaned="true">
    <package name="com.google.android.youtube" codePath="/data/app/com.google.android.youtube-74u2EpckFWqZkZREQqx5XA==" nativeLibraryPath="/dat
a/app/com.google.android.youtube-74u2EpckFWqZkZREQqx5XA==/lib" primaryCpuAbi="arm64-v8a" publicFlags="811318981" privateFlags="524288" ft="16ff
239c188" it="11e8f7d4c00" ut="16ff239c7d3" version="1503561000" userId="10196" installer="com.android.vending" isOrphaned="true" categoryHint="
2">
    <package name="com.vzw.apnlib" codePath="/product/app/VZWAPNLib" nativeLibraryPath="/product/app/VZWAPNLib/lib" publicFlags="805879365" pri
vateFlags="524288" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="17" userId="10186" isOrphaned="true">
    <package name="com.android.internal.display.cutout.emulation.corner" codePath="/product/overlay/DisplayCutoutEmulationCorner" nativeLibrary
Path="/product/overlay/DisplayCutoutEmulationCorner/lib" publicFlags="537443905" privateFlags="524288" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11
e8f7d4c00" version="1" userId="10060" isOrphaned="true">
    <package name="com.radicalapps.cyberdust" codePath="/data/app/com.radicalapps.cyberdust-rtk9Pe4GGUZS2zqlnx5GKQ==" nativeLibraryPath="/data/
app/com.radicalapps.cyberdust-rtk9Pe4GGUZS2zqlnx5GKQ==/lib" primaryCpuAbi="arm64-v8a" publicFlags="810073668" privateFlags="0" ft="16ff2428f70"
 it="16ff2429163" ut="16ff2429163" version="2887" userId="10225" installer="com.android.vending" categoryHint="4">
    <package name="com.google.android.ext.services" codePath="/system/priv-app/GoogleExtServicesPrebuilt" nativeLibraryPath="/system/priv-app/G
oogleExtServicesPrebuilt/lib" publicFlags="805879365" privateFlags="8" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11e8f7d4c00" version="291900200" u
serId="10070" isOrphaned="true">
    <package name="com.android.internal.display.cutout.emulation.double" codePath="/product/overlay/DisplayCutoutEmulationDouble" nativeLibrary
Path="/product/overlay/DisplayCutoutEmulationDouble/lib" publicFlags="537443905" privateFlags="524288" ft="11e8f7d4c00" it="11e8f7d4c00" ut="11
e8f7d4c00" version="1" userId="10024" isOrphaned="true">
```

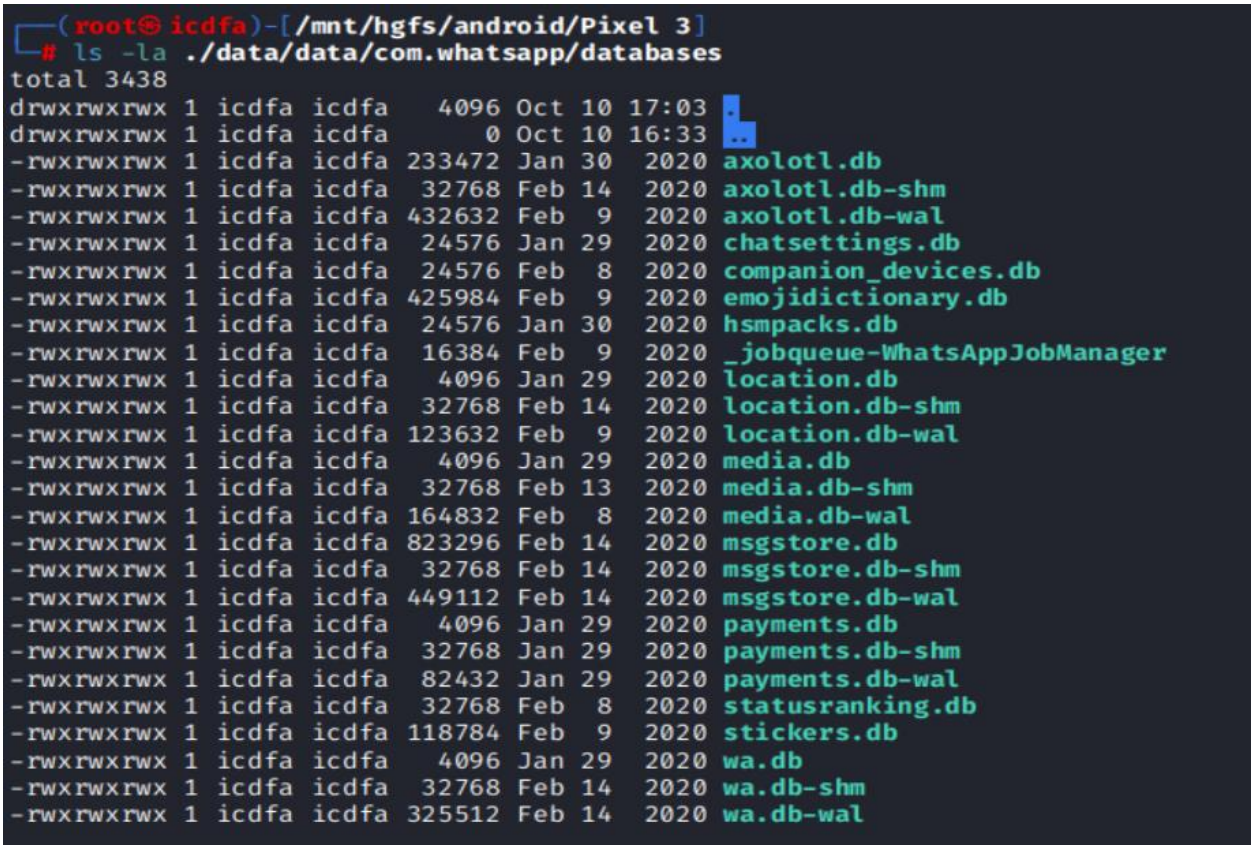
Analysis of packages.xml identified numerous installed applications on the device. Relevant applications included WhatsApp, Kik, Twitter, Google Dialer, and Google Maps. The package records confirmed that these applications were installed and configured on the device at the time of acquisition. A total of 291 package entries were identified within the package management database.

Installing Applications

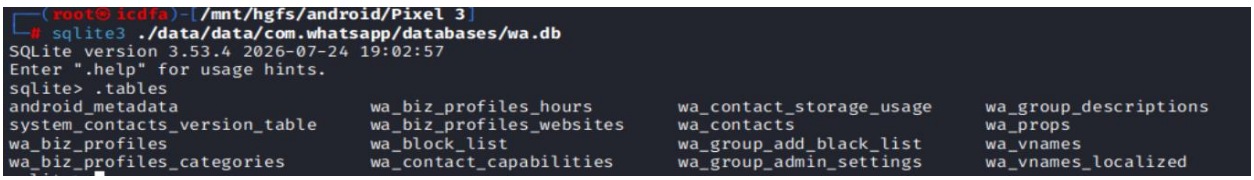
```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# grep -iE "whatsapp|telegram|textnow|kik|maps|photos|calendar|contacts|dialer|signal|skype|twitter|messenger|tiktok" ./data/system/packages.
xml
        <item name="com.google.android.apps.photos.permission.C2D_MESSAGE" package="com.google.android.apps.photos" protection="2" />
        <item name="com.google.android.calendar.permission.C2D_MESSAGE" package="com.google.android.calendar" protection="2" />
        <item name="com.google.android.googleapps.permission.GOOGLE_AUTH.sitemaps" package="com.google.android.gsf" protection="2" />
        <item name="android.permission.READ_CALENDAR" package="android" protection="1" />
        <item name="android.permission.SIGNAL_PERSISTENT_PROCESSES" package="android" protection="50" />
        <item name="kik.android.permission.CONTACT" package="kik.android" protection="2" />
        <item name="org.telegram.messenger.permission.MAPS_RECEIVE" package="org.telegram.messenger" protection="2" />
        <item name="com.whatsapp.permission.BROADCAST" package="com.whatsapp" protection="2" />
        <item name="com.skype.raider.permission.ALARM_MANAGER" package="com.skype.raider" protection="18" />
        <item name="com.google.android.gms.permission.CONTACTS_SYNC_DELEGATION" package="com.google.android.gms" protection="2" />
        <item name="android.permission.NETWORK_SIGNAL_STRENGTH_WAKEUP" package="android" protection="18" />
        <item name="com.silentcircle.silentcontacts.permission.WRITE" package="com.silentcircle.silentphone" protection="2" />
        <item name="android.permission.WRITE_CONTACTS" package="android" protection="1" />
        <item name="com.whatsapp.permission.REGISTRATION" package="com.whatsapp" protection="2" />
        <item name="com.google.android.apps.maps.permission.PREFETCH" package="com.google.android.apps.maps" protection="2" />
        <item name="com.skype.raider.permission.RECEIVE_ADM_MESSAGE" package="com.skype.raider" protection="2" />
        <item name="com.google.android.apps.maps.permission.C2D_MESSAGE" package="com.google.android.apps.maps" protection="2" />
        <item name="android.permission.WRITE_CALENDAR" package="android" protection="1" />
        <item name="com.twitter.android.permission.READ_DATA" package="com.twitter.android" protection="2" />
        <item name="com.google.android.apps.maps.permission.TIMELINE_POST_VISIT_BADGE" package="com.google.android.apps.maps" protection="2" />
        <item name="com.silentcircle.silentcontacts.permission.READ" package="com.silentcircle.silentphone" protection="2" />
        <item name="com.skype.raider.permission.ACCOUNT" package="com.skype.raider" protection="2" />
        <item name="com.whatsapp.permission.MIGRATION_CONTENT_PROVIDER" package="com.whatsapp" protection="2" />
        <item name="com.twitter.android.permission.RESTRICTED" package="com.twitter.android" protection="2" />
        <item name="com.whatsapp.sticker.READ" package="com.whatsapp" />
        <item name="com.whatsapp.permission.MAPS_RECEIVE" package="com.whatsapp" protection="2" />
        <item name="android.permission.SEND_CALL_LOG_CHANGE" package="com.android.providers.contacts" protection="18" />
        <item name="com.skype.raider.permission.C2D_MESSAGE" package="com.skype.raider" protection="2" />
        <item name="android.permission.READ_CONTACTS" package="android" protection="1" />
        <item name="com.google.android.calendar.permission.READ_OOBE" package="com.google.android.calendar" protection="2" />
    <package name="com.skype.raider" codePath="/data/app/com.skype.raider-T-lF3_kJap3DnQQzr3UxEw==" nativeLibraryPath="/data/app/com.skype.raid
er-T-lF3_kJap3DnQQzr3UxEw==/lib" primaryCpuAbi="arm64-v8a" publicFlags="810040900" privateFlags="0" ft="16ff2454aa8" it="16ff245509a" ut="16ff2
45509a" version="1250088185" userId="10236" installer="com.android.vending" categoryHint="4">
            <item name="com.skype.raider.permission.RECEIVE_ADM_MESSAGE" granted="true" flags="0" />
            <item name="com.skype.raider.permission.ACCOUNT" granted="true" flags="0" />
            <item name="com.skype.raider.permission.C2D_MESSAGE" granted="true" flags="0" />
```

Application	Package Name
WhatsApp	com.whatsapp

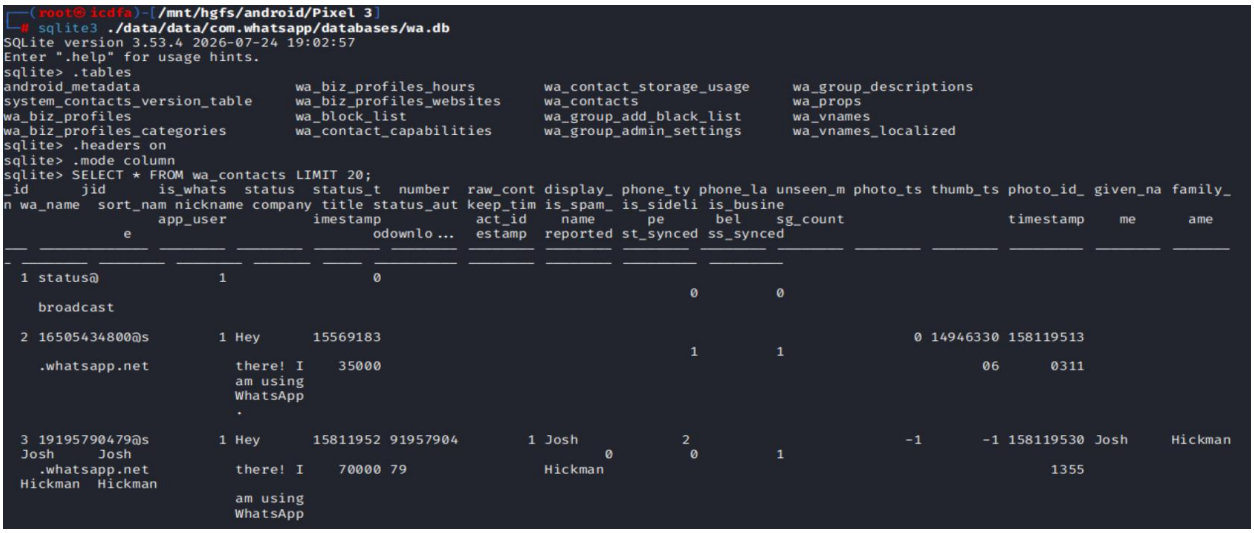
Kik	kik.android
Google Maps	com.google.android.apps.maps
Google Dialer	com.google.android.dialer
Twitter	com.twitter.android
Skype	com.skype.raider
Telegram	org.telegram.messenger



WhatsApp databases identified within the application data directory.



WhatsApp database tables recovered from wa.db.

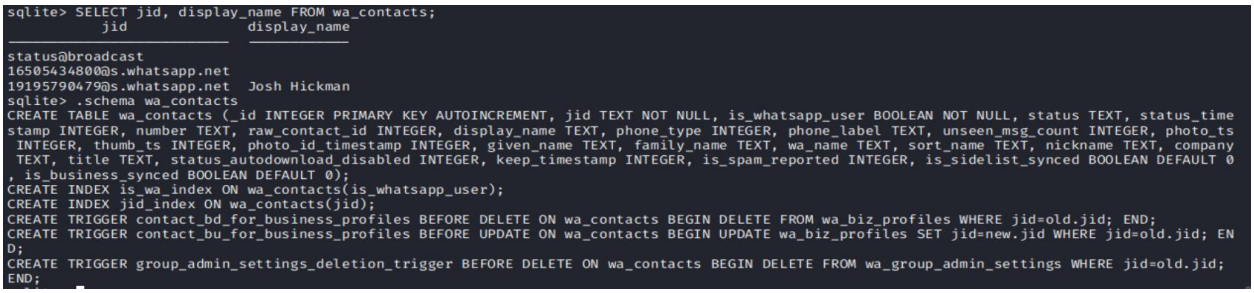


```
┌──(root㉿icdfa)-[/mnt/hgfs/android/Pixel 3]
└─# sqlite3 ./data/data/com.whatsapp/databases/wa.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .tables
android_metadata                wa_biz_profiles_hours       wa_contact_storage_usage    wa_group_descriptions
system_contacts_version_table   wa_biz_profiles_websites     wa_contacts                 wa_props
wa_biz_profiles                 wa_block_list               wa_group_add_black_list     wa_vnames
wa_biz_profiles_categories      wa_contact_capabilities     wa_group_admin_settings     wa_vnames_localized
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT * FROM wa_contacts LIMIT 20;
_id      jid     is_whats  status  status_t  number  raw_cont display_ phone_ty phone_la unseen_m photo_ts thumb_ts photo_id_ given_na family_
n wa_name  sort_nam nickname company title status_aut keep_tim is_spam_ is_sideli is_busine
                app_user          imestamp           act_id    name      pe      bel     sg_count                  timestamp    me       ame
             e                           odownlo ...  estamp  reported st_synced ss_synced
___ _________ ________ ________ _______ _____ _________ ________ ________ ________ ________ ________ ________ _________ ________ _______
- ________ ________ ________ _______ ____ _________ ________ ________ ________ ________
  1 status@             1                 0
                                                                                  0         0
    broadcast

  2 16505434800@s       1 Hey       15569183                                                      0 14946330 158119513
                                                                                  1         1
    .whatsapp.net         there! I    35000                                                              06       0311
                          am using
                          WhatsApp
                          .

  3 19195790479@s       1 Hey       15811952 91957904        1 Josh            2                       -1       -1 158119530 Josh     Hickman
  Josh     Josh                                                        0         0         1
    .whatsapp.net         there! I    70000 79                 Hickman                                                1355
  Hickman  Hickman
                          am using
                          WhatsApp
```

Analysis of wa.db identified WhatsApp contact artefacts stored within the wa_contacts table. Recovered records included WhatsApp identifiers (JIDs), contact names, telephone numbers, and profile status information. The examination confirmed active use of WhatsApp and the presence of stored contact information on the device.



```
sqlite> SELECT jid, display_name FROM wa_contacts;
          jid             display_name
_________________  ____________
status@broadcast
16505434800@s.whatsapp.net
19195790479@s.whatsapp.net  Josh Hickman
sqlite> .schema wa_contacts
CREATE TABLE wa_contacts (_id INTEGER PRIMARY KEY AUTOINCREMENT, jid TEXT NOT NULL, is_whatsapp_user BOOLEAN NOT NULL, status TEXT, status_time
stamp INTEGER, number TEXT, raw_contact_id INTEGER, display_name TEXT, phone_type INTEGER, phone_label TEXT, unseen_msg_count INTEGER, photo_ts
 INTEGER, thumb_ts INTEGER, photo_id_timestamp INTEGER, given_name TEXT, family_name TEXT, wa_name TEXT, sort_name TEXT, nickname TEXT, company
 TEXT, title TEXT, status_autodownload_disabled INTEGER, keep_timestamp INTEGER, is_spam_reported INTEGER, is_sidelist_synced BOOLEAN DEFAULT 0
, is_business_synced BOOLEAN DEFAULT 0);
CREATE INDEX is_wa_index ON wa_contacts(is_whatsapp_user);
CREATE INDEX jid_index ON wa_contacts(jid);
CREATE TRIGGER contact_bd_for_business_profiles BEFORE DELETE ON wa_contacts BEGIN DELETE FROM wa_biz_profiles WHERE jid=old.jid; END;
CREATE TRIGGER contact_bu_for_business_profiles BEFORE UPDATE ON wa_contacts BEGIN UPDATE wa_biz_profiles SET jid=new.jid WHERE jid=old.jid; EN
D;
CREATE TRIGGER group_admin_settings_deletion_trigger BEFORE DELETE ON wa_contacts BEGIN DELETE FROM wa_group_admin_settings WHERE jid=old.jid;
END;
```

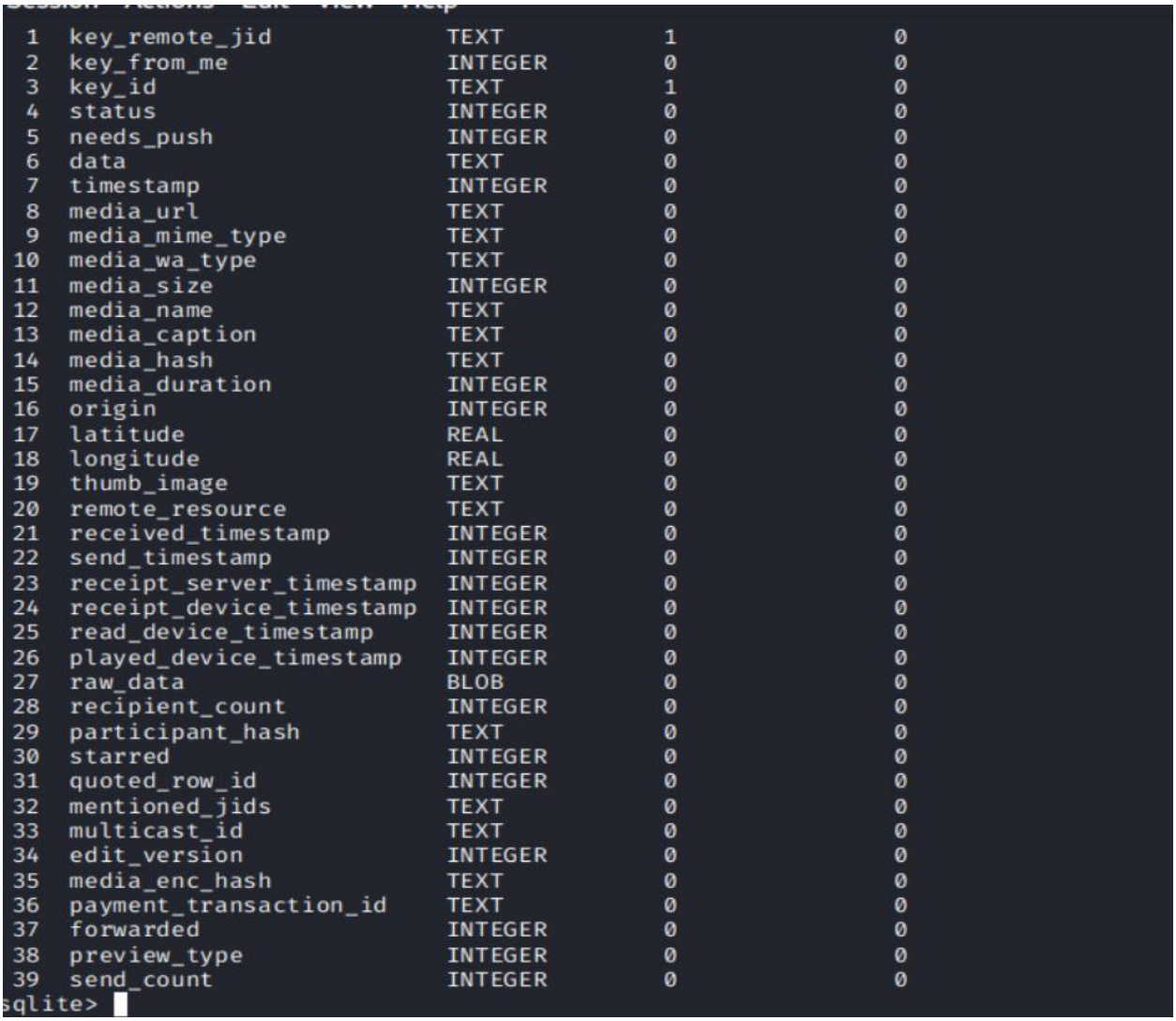
Examination of wa.db identified contact information stored within the wa_contacts table. WhatsApp contact records included user identifiers (JIDs) and associated display names. One recovered contact was identified as *Josh Hickman* (19195790479@s.whatsapp.net). The database also contained the device owner's WhatsApp account identifier (16505434800@s.whatsapp.net). These findings confirm active WhatsApp usage and storage of contact information on the device.

```
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT _id, raw_string FROM jid LIMIT 20;
_id         raw_string
--  ------------------------------
2   0@s.whatsapp.net
3   16505434800@s.whatsapp.net
5   19195790479.0:0@s.whatsapp.net
4   19195790479@s.whatsapp.net
1   status@broadcast
sqlite> SELECT * FROM jid LIMIT 20;
_id  user         server         agent  device  type  raw_string
---  -----------  -------------  -----  ------  ----  ------------------------------
1    status       broadcast          0       0     5  status@broadcast
2    0            s.whatsapp.net     0       0     7  0@s.whatsapp.net
3    16505434800  s.whatsapp.net     0       0     0  16505434800@s.whatsapp.net
4    19195790479  s.whatsapp.net     0       0     0  19195790479@s.whatsapp.net
5    19195790479  s.whatsapp.net     0       0    17  19195790479.0:0@s.whatsapp.net
sqlite>
```

```
sqlite> SELECT _id, user, raw_string FROM jid;
_id     user          raw_string
---  -----------  ------------------------------
1    status       status@broadcast
2    0            0@s.whatsapp.net
3    16505434800  16505434800@s.whatsapp.net
4    19195790479  19195790479@s.whatsapp.net
5    19195790479  19195790479.0:0@s.whatsapp.net
```

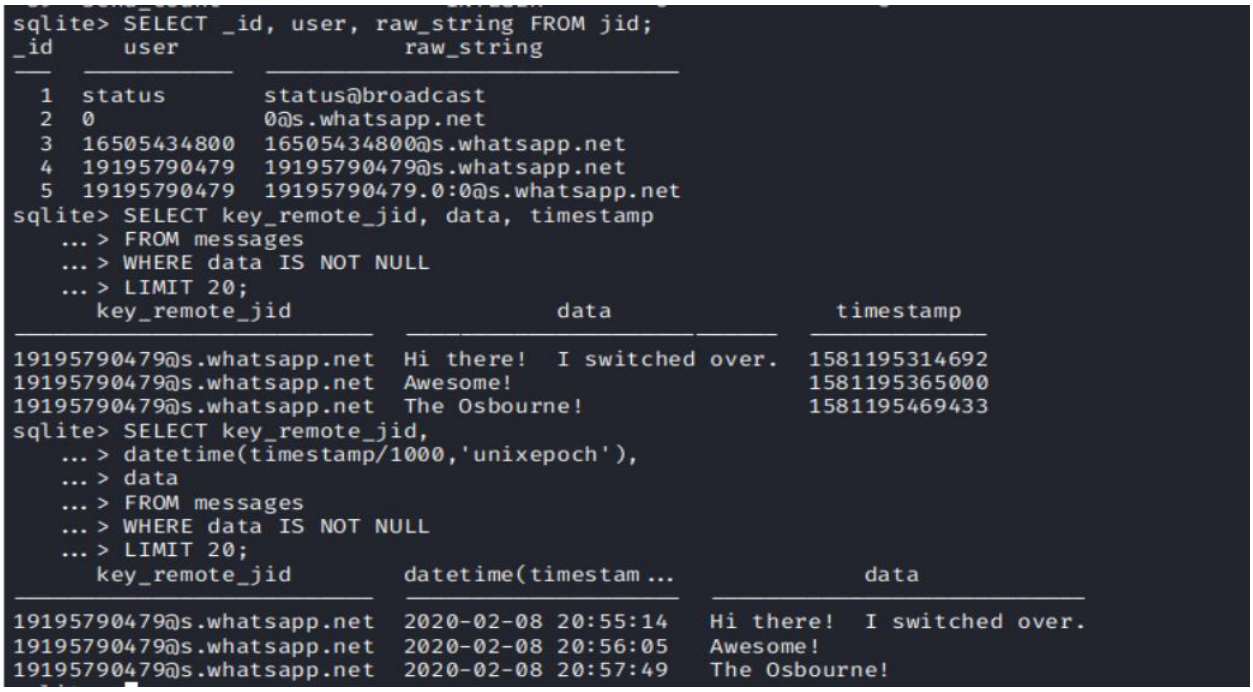
WhatsApp JIDs recovered from the jid table in msgstore.db.

Analysis of WhatsApp's msgstore.db database identified multiple WhatsApp JIDs. Recovered account identifiers included 16505434800@s.whatsapp.net and 19195790479@s.whatsapp.net. These records demonstrate that the application contained communication relationships associated with these WhatsApp accounts.



```
 1  key_remote_jid            TEXT          1             0
 2  key_from_me               INTEGER       0             0
 3  key_id                    TEXT          1             0
 4  status                    INTEGER       0             0
 5  needs_push                INTEGER       0             0
 6  data                      TEXT          0             0
 7  timestamp                 INTEGER       0             0
 8  media_url                 TEXT          0             0
 9  media_mime_type           TEXT          0             0
10  media_wa_type             TEXT          0             0
11  media_size                INTEGER       0             0
12  media_name                TEXT          0             0
13  media_caption             TEXT          0             0
14  media_hash                TEXT          0             0
15  media_duration            INTEGER       0             0
16  origin                    INTEGER       0             0
17  latitude                  REAL          0             0
18  longitude                 REAL          0             0
19  thumb_image               TEXT          0             0
20  remote_resource           TEXT          0             0
21  received_timestamp        INTEGER       0             0
22  send_timestamp            INTEGER       0             0
23  receipt_server_timestamp  INTEGER       0             0
24  receipt_device_timestamp  INTEGER       0             0
25  read_device_timestamp     INTEGER       0             0
26  played_device_timestamp   INTEGER       0             0
27  raw_data                  BLOB          0             0
28  recipient_count           INTEGER       0             0
29  participant_hash          TEXT          0             0
30  starred                   INTEGER       0             0
31  quoted_row_id             INTEGER       0             0
32  mentioned_jids            TEXT          0             0
33  multicast_id              TEXT          0             0
34  edit_version              INTEGER       0             0
35  media_enc_hash            TEXT          0             0
36  payment_transaction_id    TEXT          0             0
37  forwarded                 INTEGER       0             0
38  preview_type              INTEGER       0             0
39  send_count                INTEGER       0             0
sqlite>
```

Structure of the WhatsApp messages table showing message content, timestamps, media and location fields



WhatsApp message records recovered from msgstore.db, showing contact identifiers, timestamps and message content.

Analysis of the WhatsApp database (msgstore.db) identified message communications associated with the contact 19195790479@s.whatsapp.net. Several message records were recovered, including:

- “Hi there! I switched over.”
- “Awesome!”
- “The Osbourne!”

Timestamp analysis showed these messages were exchanged on **8 February 2020** between **20:55:14** and **20:57:49**. The recovery of message content, contact identifiers and timestamps demonstrates that meaningful communication artefacts were successfully extracted from the WhatsApp application database.

Timestamp	Contact	Message
2020-02-08 20:55:14	19195790479@s.whatsapp.net	Hi there! I switched over.
2020-02-08 20:56:05	19195790479@s.whatsapp.net	Awesome!
2020-02-08 20:57:49	19195790479@s.whatsapp.net	The Osbourne!

WhatsApp Call Logs



```
sqlite> SELECT * FROM call_log LIMIT 20;
_id jid_row_id from_me               call_id                transaction_id   timestamp   video_call duration call_result bytes_transferred
___ __________ _______ ____________________________________ ______________ _____________ __________ ________ ___________ _________________
  1          4       0 call:5B7EAA540D7ED444773C15A167D1CFDB             -1 1581195562000          1       86           5          22302020
  2          4       1 call:E2D103295A0D7DE5F930B7DE6C971F1C             -1 1581195700411          0       71           5            183698
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT * FROM call_log;
_id jid_row_id from_me               call_id                transaction_id   timestamp   video_call duration call_result bytes_transferred
___ __________ _______ ____________________________________ ______________ _____________ __________ ________ ___________ _________________
  1          4       0 call:5B7EAA540D7ED444773C15A167D1CFDB             -1 1581195562000          1       86           5          22302020
  2          4       1 call:E2D103295A0D7DE5F930B7DE6C971F1C             -1 1581195700411          0       71           5            183698
sqlite> SELECT datetime(timestamp/1000,'unixepoch')
   ...> FROM call_log;
datetime(timestam...
___________________
2020-02-08 20:59:22
2020-02-08 21:01:40
```

WhatsApp call records showing recovered timestamps and call durations.

Type	Date & Time	Duration
Video Call	2020-02-08 20:59:22	86 sec
Voice Call	2020-02-08 21:01:40	71 sec

Google Maps location artefact recovered from gmm_myplaces.db showing stored coordinates and timestamp information.



```
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT * FROM sync_item LIMIT 20;
corpus  key_string    timestamp     merge_key  feature_fprint  latitude  longitude  is_local   sync_item
______  ____________  ____________  _________  ______________  ________  _________  ________  _________
     8  1:0           1554407343604          0                  35734602  -78636654          0  ^R^C1:02•^A
                                                                                                ^D^H^A^P

sqlite> SELECT corpus,key_string,timestamp,latitude,longitude
   ...> FROM sync_item
   ...> LIMIT 20;
corpus  key_string    timestamp     latitude  longitude
______  ____________  ____________  ________  _________
     8  1:0           1554407343604  35734602  -78636654
sqlite> SELECT datetime(timestamp/1000,'unixepoch')
   ...> FROM sync_item;
datetime(timestam...
___________________
2019-04-04 19:49:03
```

Analysis of Google Maps artefacts identified location-related records within gmm_myplaces.db. A recovered record contained timestamp and geographic coordinate information (35.734602, -78.636654), indicating the storage of Maps-related location data. The artefact demonstrates that Google Maps maintained historical user-location information that may assist in reconstructing device activity and travel patterns.

Timestamp	Latitude	Longitude
1554407343604	35.734602	-78.636654

Google Photos

Analysis of Google Photos artefacts identified multiple media records within the gphotos-1.db database. The media table contained timestamps associated with photo and video assets,

demonstrating that the application maintained historical records of media content accessed or stored by the user. Recovered records included media timestamps between September 2019 and February 2020. The artefacts examined showed no evidence that the displayed items had been deleted or archived (is_deleted = 0, is_archived = 0).

The recovered timestamps indicate user interaction with media content between September 2019 and February 2020.

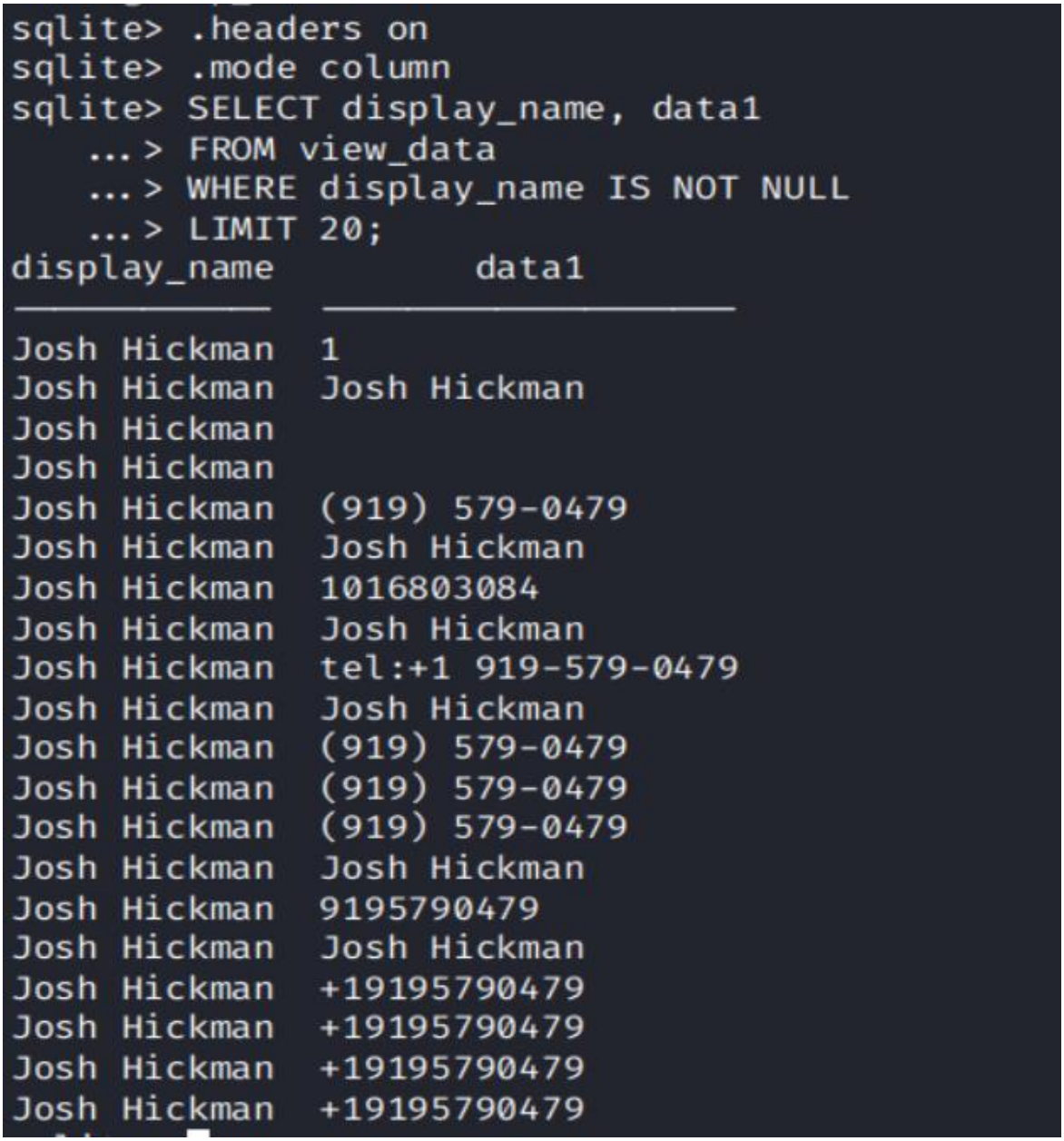
```
sqlite> SELECT _id,
   ...> datetime(utc_timestamp/1000,'unixepoch'),
   ...> type,
   ...> is_deleted,
   ...> is_archived
   ...> FROM media
   ...> LIMIT 20;
```

_id	datetime(utc_time ...	type	is_deleted	is_archived
1	2020-02-01 02:25:44	1	0	0
2	2019-09-17 12:26:37	1	0	0
3	2020-02-01 02:15:54	1	0	0
4	2020-02-01 01:43:23	1	0	0
10	2020-02-01 20:16:29	1	0	0
11	2020-02-01 19:53:50	1	0	0
13	2020-02-01 19:34:44	1	0	0
14	2020-02-01 19:22:07	1	0	0
16	2020-02-01 19:15:07	1	0	0
17	2020-02-01 19:14:29	1	0	0
18	2020-02-01 19:10:18	1	0	0
19	2020-02-01 19:07:05	1	0	0
20	2020-02-01 18:58:14	1	0	0
40	2020-02-02 00:47:45	1	0	0
42	2020-02-04 02:30:34	1	0	0
43	2020-02-04 02:30:26	1	0	0
44	2020-02-04 02:02:27	1	0	0
86	2020-02-09 17:55:53	2	0	0
87	2020-02-08 20:57:08	1	0	0
88	2020-02-08 20:35:37	1	0	0

Google Photos media records recovered from gphotos-1.db showing timestamps and media status information.

Contacts

Analysis of the Android contacts database (contacts2.db) identified stored contact records on the device. Contact information recovered included the contact **Josh Hickman** together with associated telephone numbers including **(919) 579-0479** and **+19195790479**. These artefacts demonstrate that contact information was stored locally and available to communication applications installed on the device.



```
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT display_name, data1
   ...> FROM view_data
   ...> WHERE display_name IS NOT NULL
   ...> LIMIT 20;
display_name         data1
_____________  ____________________
Josh Hickman  1
Josh Hickman  Josh Hickman
Josh Hickman
Josh Hickman
Josh Hickman  (919) 579-0479
Josh Hickman  Josh Hickman
Josh Hickman  1016803084
Josh Hickman  Josh Hickman
Josh Hickman  tel:+1 919-579-0479
Josh Hickman  Josh Hickman
Josh Hickman  (919) 579-0479
Josh Hickman  (919) 579-0479
Josh Hickman  (919) 579-0479
Josh Hickman  Josh Hickman
Josh Hickman  9195790479
Josh Hickman  Josh Hickman
Josh Hickman  +19195790479
Josh Hickman  +19195790479
Josh Hickman  +19195790479
Josh Hickman  +19195790479
```

Contact records recovered from contacts2.db showing stored contact details and associated telephone numbers.

Contact	Number
Josh Hickman	(919) 579-0479
Josh Hickman	+19195790479

Challenges Faced Section

A significant challenge encountered during the investigation was environment setup. The supplied forensic environment could not be successfully executed in VMware, requiring migration to VirtualBox before analysis could begin. This process increased preparation time and delayed evidence acquisition. Additional challenges included navigating large application datasets and identifying the most relevant artefacts among extensive application cache, configuration and database files.

Conclusion

The forensic examination of the Google Pixel 3 Android 10 image successfully identified a broad range of evidential artefacts from both system and application sources. Device profiling established the handset configuration, while account, Wi-Fi and Bluetooth artefacts provided evidence of user activity and device usage. Application analysis recovered WhatsApp communications and call records, Google Maps location data, Google Photos media information and stored contact details. The combined artefacts enabled reconstruction of user activity and demonstrated the evidential value of Android application databases during mobile forensic investigations.

Appendix – Evidence Screenshots

Figure 1 – SH***6 Verification
Figure 2 – Device***ild Information
Figure 3 – User ***ounts
Figure 4 – User Profiles
F***re 5 – Wi-Fi Artefacts
Figure 6 ***luetooth Artefacts
Figure 7 – In***lled Applications
Figure 8 – Wha***pp Contacts
Figure 9 – WhatsApp ***sages
Figure 10 – WhatsApp Calls***gure 11 – Google Maps Artefact
F***re 12 – Google Photos Artefact
Figure 13 – Contacts Artefact